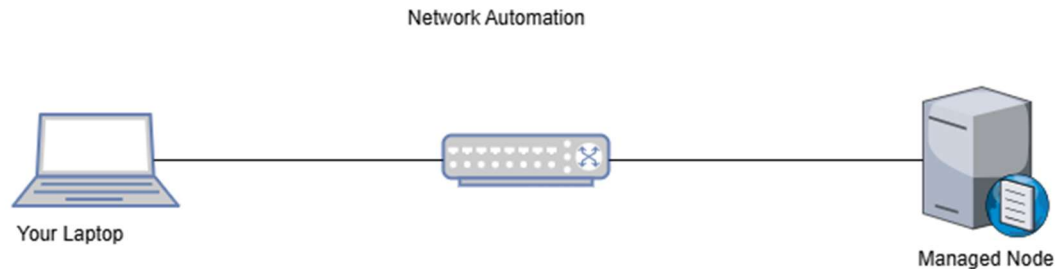


Lab 0 – Copying Files between Windows and Linux

In this lab you will work on both Windows and Linux. It shows how to copy your code files to the Linux server where they are executed.



Aim:

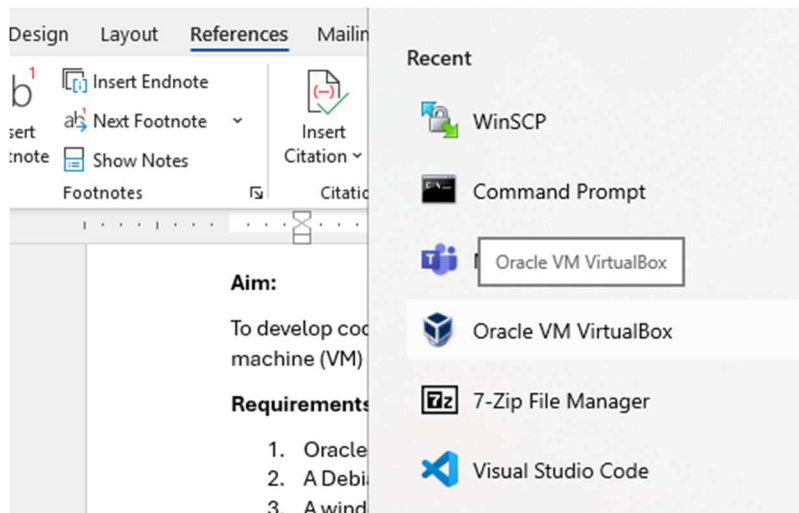
To develop code on Windows using the vscode application and then copy the code to a Linux virtual machine (VM) using winscp.

Requirements:

1. Oracle VM Virtualbox application
2. A Debian 11¹ VM image
3. A windows host computer with vscode application
4. Winscp application installed on the Windows host

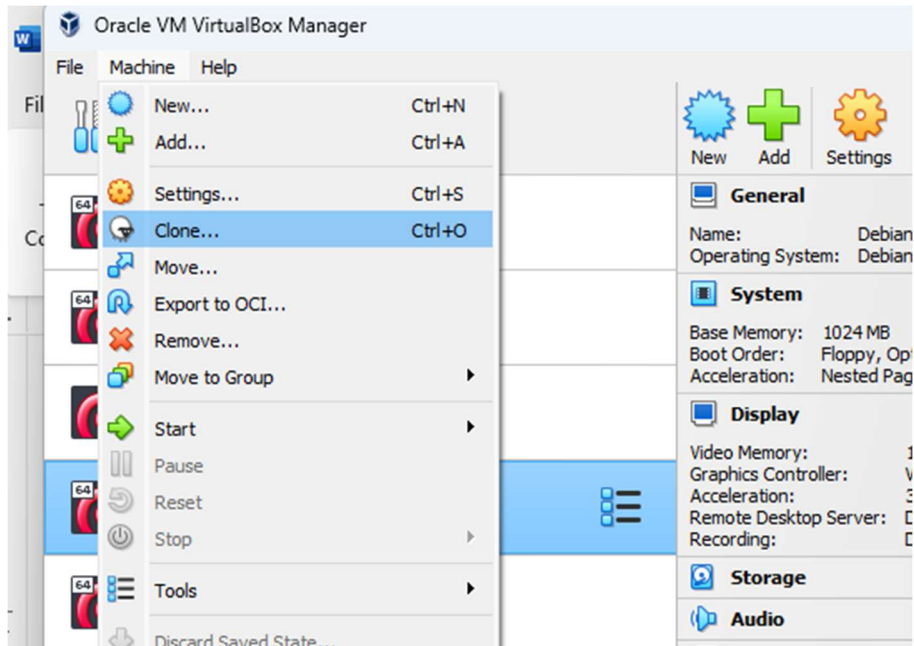
Part 1: VM preparation – Do this on you Windows Host

1. Open Oracle Virtualbox (or any other Hypervisor)

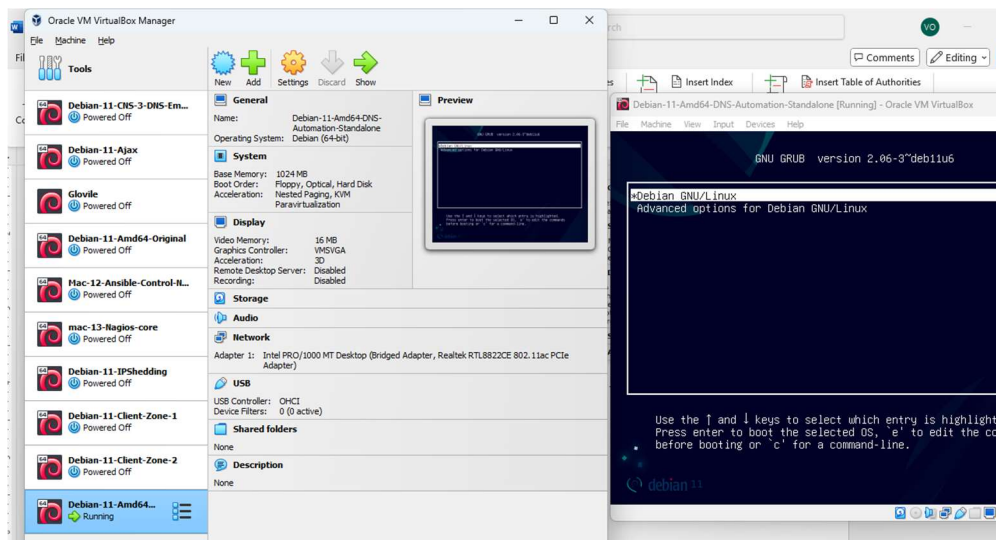


¹ Any Linux distribution will work

2. Create a fresh VM² by cloning from a Debian image

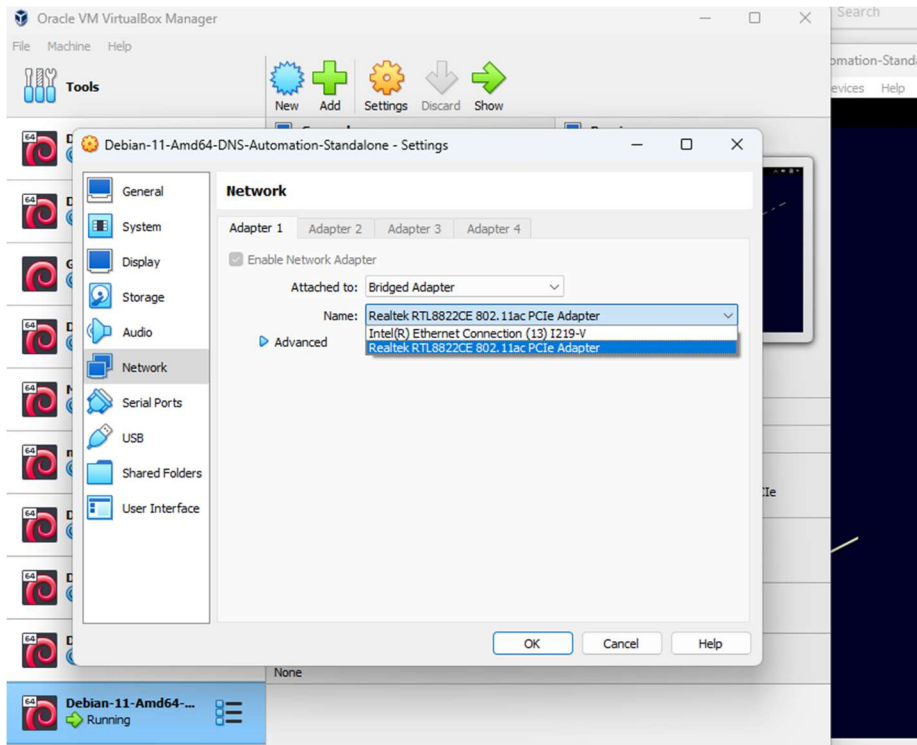


3. Startup the new VM



4. To ensure that your VM and host computer can communicate, choose Bridged Adapter under Network Settings. Then select the applicable interface e.g. WiFi or Fixed Ethernet

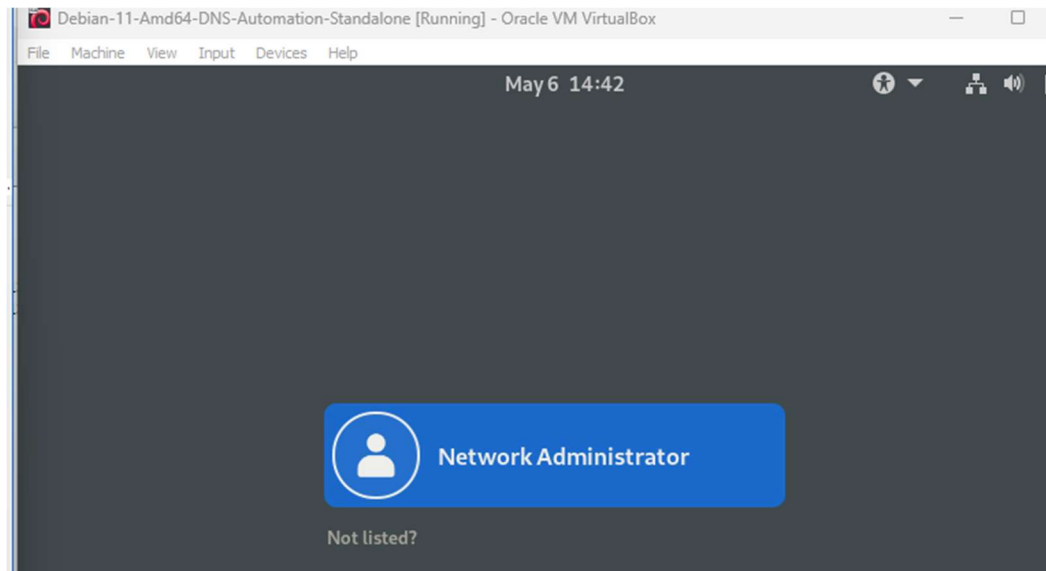
² You can use an existing VM



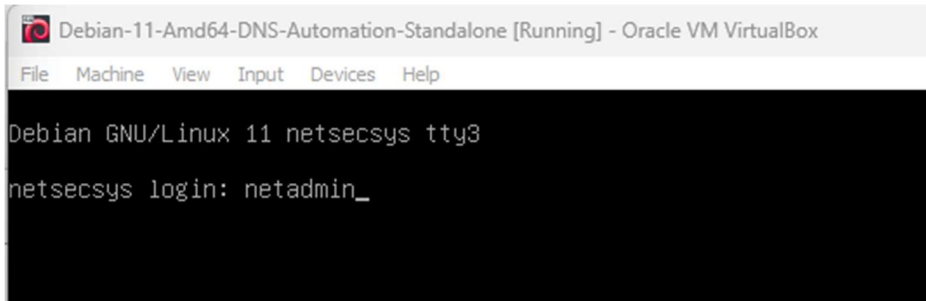
Part 2: Prepare the Debian VM – Do this on the Debian VM

5. Login via GUI or CLI. We have created a user named Network Administrator with login Name: **netadmin**

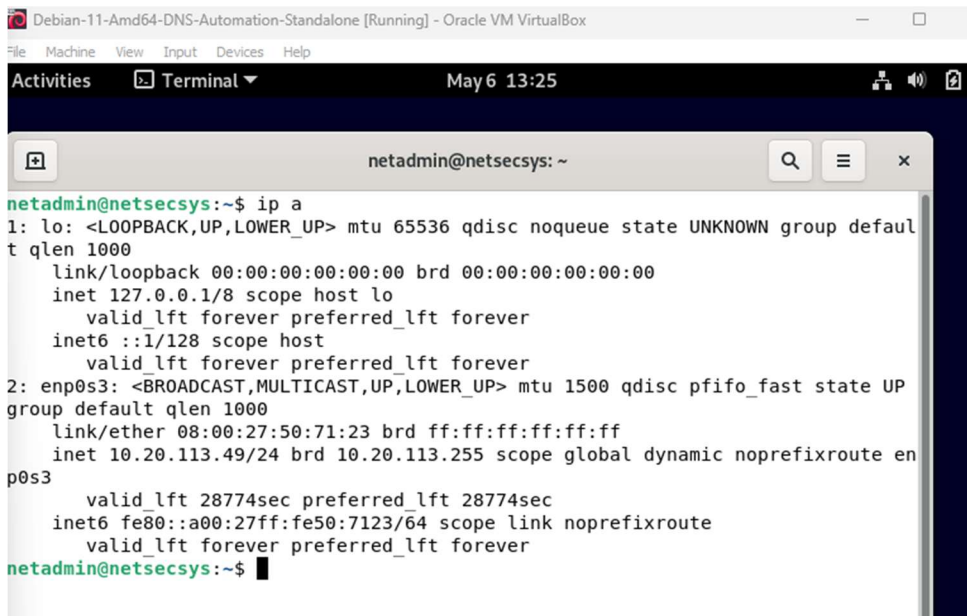
GUI



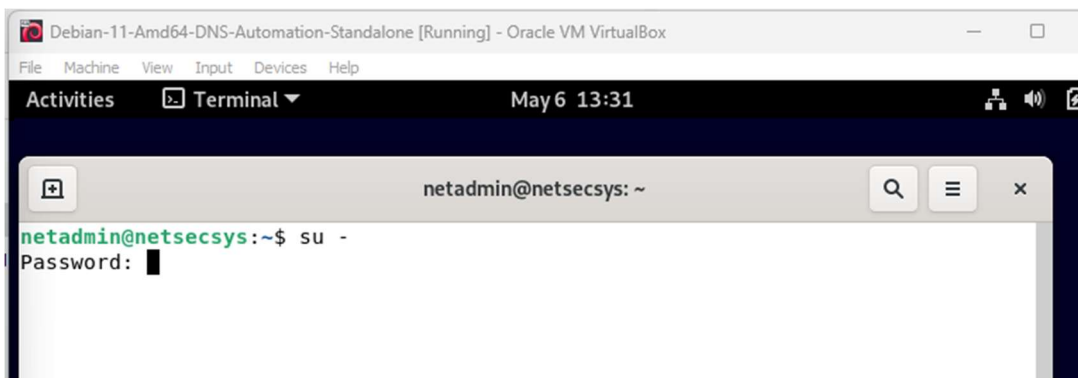
CLI



6. Login to the Debian VM, open a terminal and verify the IP address using the command:
ip a³



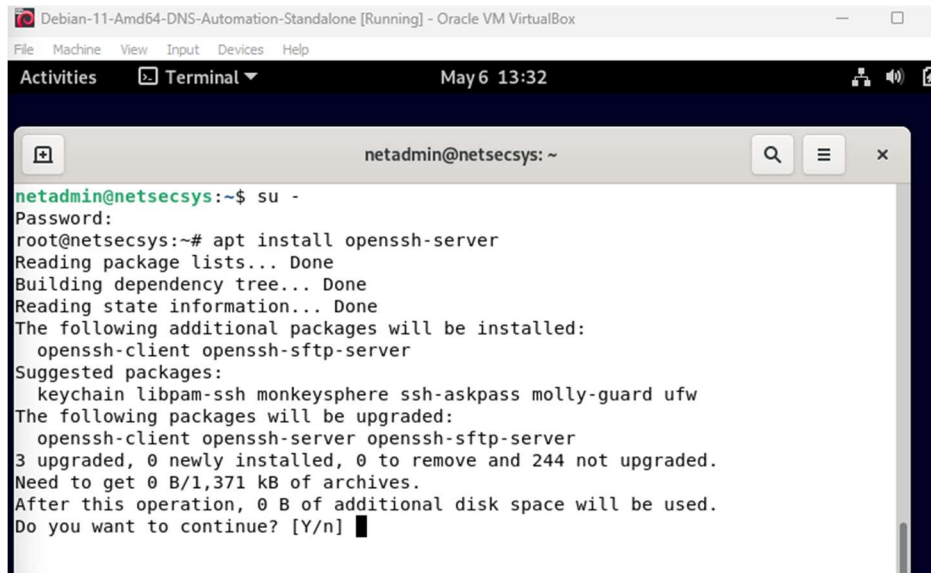
7. Login as full root on the terminal



8. Install SSH server using the command

³ In this case the IP address is 10.20.113.49

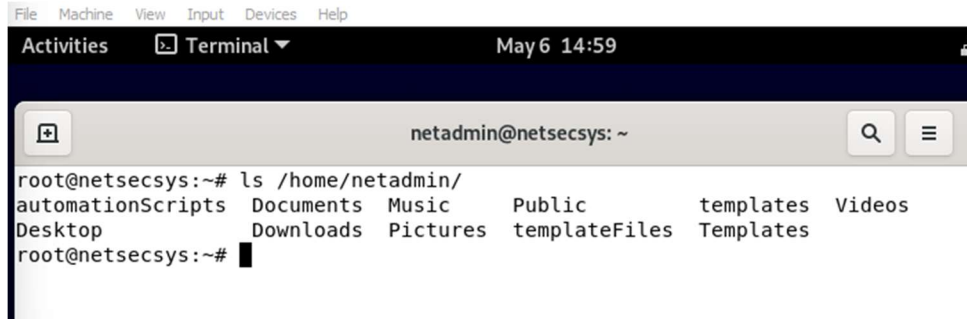
apt install openssh-server



```
Debian-11-Amd64-DNS-Automation-Standalone [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal May 6 13:32

netadmin@netsecsys: ~
netadmin@netsecsys:~$ su -
Password:
root@netsecsys:~# apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  openssh-client openssh-sftp-server
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard ufw
The following packages will be upgraded:
  openssh-client openssh-server openssh-sftp-server
3 upgraded, 0 newly installed, 0 to remove and 244 not upgraded.
Need to get 0 B/1,371 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n]
```

9. Do this after you have completed step 16 below.
 - a. Verify that files have been copied to the Debian VM using the command:
ls /home/skyadmin



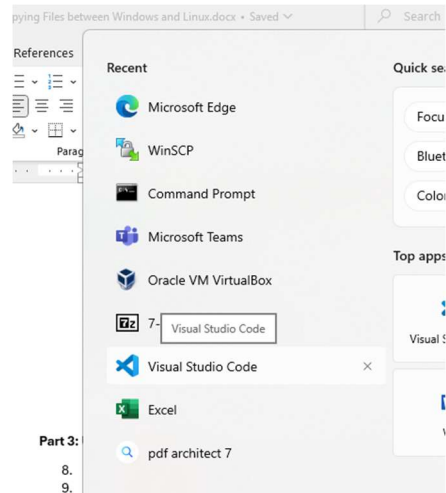
```
File Machine View Input Devices Help
Activities Terminal May 6 14:59

netadmin@netsecsys: ~
root@netsecsys:~# ls /home/netadmin/
automationScripts  Documents  Music      Public      templates  Videos
Desktop            Downloads  Pictures   templateFiles  Templates
```

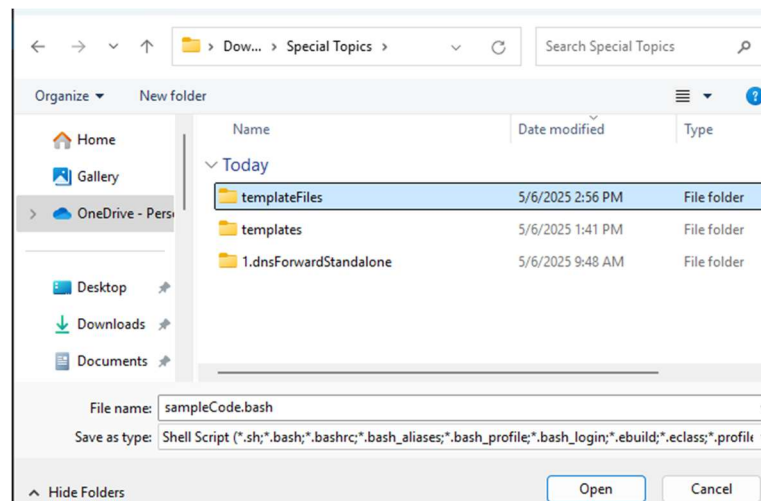
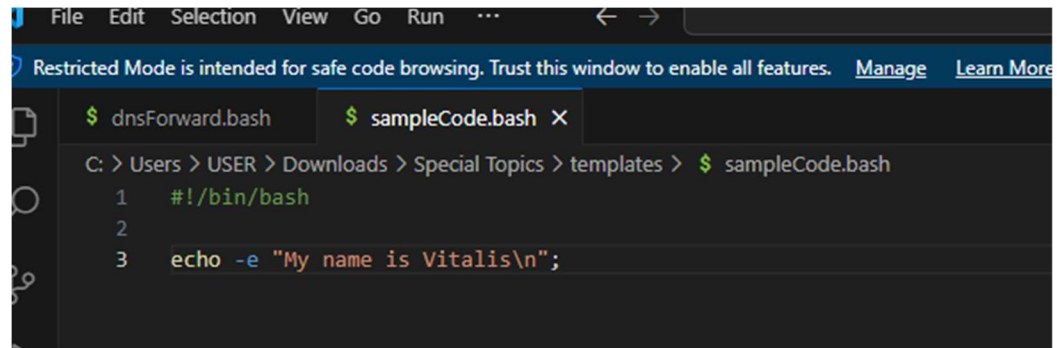
The **templateFiles** folder should be displayed.

Part 3: Using winscp application – Do this on the Windows host

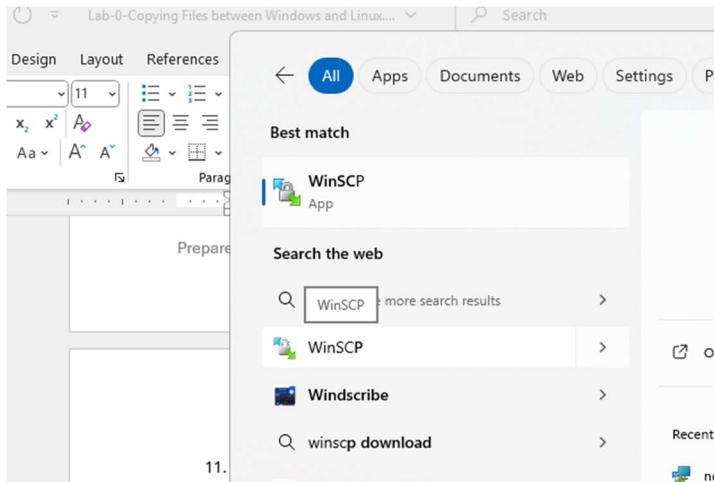
10. Download winscp from <https://winscp.net/eng/download.php>
11. Download vscode or another IDE you like to use for coding. Get vscode from <https://code.visualstudio.com/download>
12. Use vscode application then create a file and write the code you want to run or upload on the Debian VM
 - a. Open vscode application



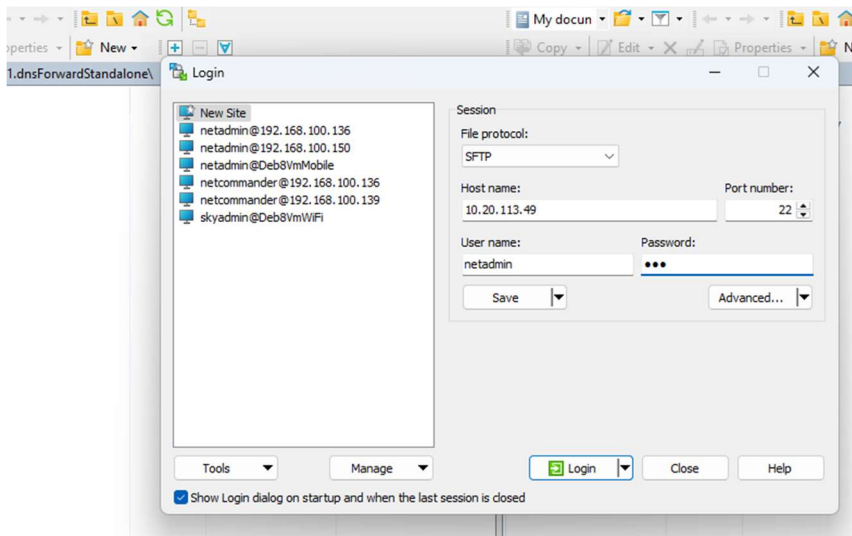
- b. Create a file named **sampleCode.bash** and save it in a directory named **templateFiles**



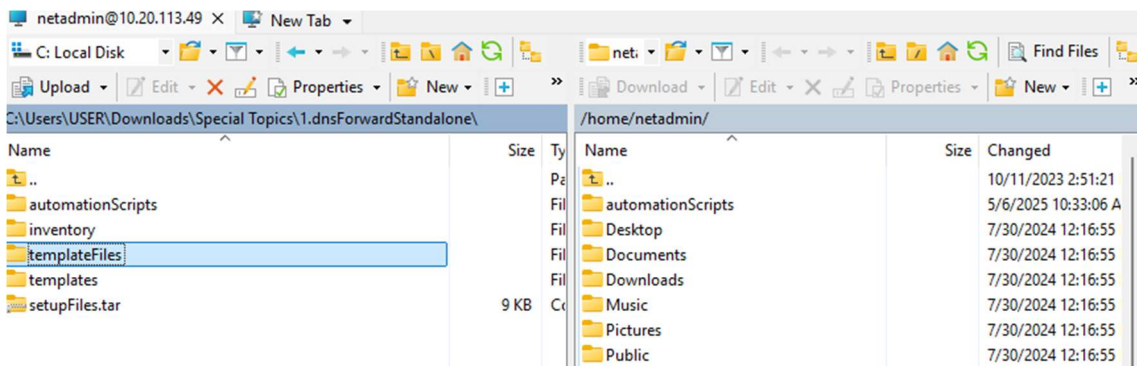
13. Open winscp



14. Login to the Debian VM by specifying the Host name (can use IP address = 10.20.113.49), User Name (netadmin) and password.



15. Once logged in, the Left hand side panel of winscp shows files on your Windows computer, while files on your Debian Linux VM appear on the Right hand side panel.



16. To copy the templateFiles folder from the Windows host machine to the Debian VM, drag-and-drop from the Left hand side panel to the right hand side panel.
17. To copy files from the Debian VM to the Windows host, drag-and-drop from the Right hand side pane to the Left hand side pane.